

Fall 2025, EE 582: Assignment 5:

Synchronous Machine Modeling

Due on Thursday, November 20th (submission in Canvas)

- 1) Develop a $qd0$ model (in per-unit) of a 325-MVA **Hydro-Turbine Generator** (book p. 220) in Simulink using basic blocks. The generator is to be connected to an ideal source (infinite bus) that supplies balanced three-phase sinusoidal voltages at rated 1pu. Assume all variables are in pu, unless otherwise specified. Assume positive direction of T_m is toward increasing the rotor speed ω_r and angle δ_r . Choose an ode solver with the settings (error tolerances and step-size limits) that you think produces trustable results and is numerically efficient. Comment on your choice.

Table 5.10-1 Hydro Turbine Generator

Rating: 325 MVA	
Line-to-line voltage: 20 kV	
Power factor: 0.85	
Poles: 64	
Speed: 112.5 r/min	
Combined inertia of generator and turbine:	
$J = 35.1 \times 10^6 \text{ J} \cdot \text{s}^2$, or $WR^2 = 833.1 \times 10^6 \text{ lbm} \cdot \text{ft}^2$ $H = 7.5 \text{ s}$	
Parameters in ohms and per unit:	
$r_s = 0.00234 \Omega, 0.0019 \text{ pu}$	
$X_{ls} = 0.1478 \Omega, 0.120 \text{ pu}$	
$X_q = 0.5911 \Omega, 0.480 \text{ pu}$	$X_d = 1.0467 \Omega, 0.850 \text{ pu}$
	$r'_{fd} = 0.00050 \Omega, 0.00041 \text{ pu}$
	$X'_{fd} = 0.2523 \Omega, 0.2049 \text{ pu}$
$r'_{kq2} = 0.01675 \Omega, 0.0136 \text{ pu}$	$r'_{kd} = 0.01736 \Omega, 0.0141 \text{ pu}$
$X'_{lkq2} = 0.1267 \Omega, 0.1029 \text{ pu}$	$X'_{lkd} = 0.1970 \Omega, 0.160 \text{ pu}$

- 2) Assume zero initial conditions and $E_{x_{fd}} = 0$. Implement a start-up transient similar to an induction machine (i.e., starting this synchronous machine from stall). Plot variables i_{as} , T_e , ω_r . Comment on the results. Explain how it would be practical to bring a large machine like this online.
- 3) Calculate the initial conditions that would correspond to an idle steady-state operation with $T_m = 0$ and excitation $E_{x_{fd}} = 1.0 \text{ pu}$, and set the initial conditions appropriately in your model. Implement the following transient study: At $t = 1 \text{ s}$, the excitation $E_{x_{fd}}$ is stepped down to 0.8 pu. Then, at $t = 7 \text{ s}$, the excitation $E_{x_{fd}}$ is stepped up to 1.2 pu. Then, at $t = 15 \text{ s}$, the mechanical torque is stepped to $T_m = 0.85 \text{ pu}$ supplying power from the mechanical system to the electrical system, and the model is continued to run till $t = 25 \text{ s}$. Plot variables i_{as} , i_{fd} , i_{kq} , T_e , ω_r , δ_r (in electrical degrees), as well as real power P_e and reactive power Q_e . Based on the transients observed, briefly explain the effect of the excitation and torque on the active and reactive power and the type of transients seen.

- 4) Now assume that the q -axis damper winding is damaged. This can be implemented by setting the resistance of this winding to a very high value, e.g., set $r_{kq} = 1$ pu . Repeat what you did in part 3).
 - 5) Now assume that the d -axis damper winding is damaged. Repeat what you did in part 3). No need to plot anything. Just make observations.
 - 6) Based on the observations in parts 4) and 5), explain the effect and the role of each of these damper windings.
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REPORTING

All reports should be typed. The assignment report should demonstrate the individual's work and include printouts of the model windows, simulation results, as well as answers, discussions, and conclusions wherever appropriate. Use graphical commands such as **hold** and **subplot** for compact and clear presentation of the figures in your report. The discussions and comments should not be long (**avoid writing unnecessarily long reports!**), but they should demonstrate your understanding of the problem.

Please include your MATLAB codes as well as screenshots of your Simulink models as part of your report. Your report should be one single pdf file.

Best of luck!